1. Introduction to the *Handbook on Public Policy and Artificial Intelligence*: vantage points for critical inquiry

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INTRODUCTION

In 2015, the US Citizenship and Immigration Services (USCIS) launched a chatbot named Emma. As USCIS explains, Emma is "a computer-generated virtual assistant who can answer your questions and even take you to the right spot on our website" both in English and Spanish (US Citizenship and Immigration Services 2018). Emma is one example of how so-called artificial intelligence technologies (AITs) have entered the public sector (and, of course, beyond). The use of AIT in software and hardware is often much more mundane, and also less visible than Emma, which "simulates" a specific human cognitive capacity relatively holistically based on computer-processed code (here: natural language processing, search for and filtering of suitable answers to a customer query). Emma presents potential and actual immigrants to the US with a friendly tone and being here to help, using the familiar visual grammar of "speech bubbles" and service-oriented structured interaction. Emma feels "easy to use" because such chatbots are so commonplace; they have become an unremarkable feature of everyday digital life, whether paying a utility bill, ordering food, or buying clothes online.

But Emma is not a chatbot of everyday life; it does not facilitate a customer relationship or help someone buy a product. Emma is the bright and shiny fascia of a repressive, opaque and violent infrastructure of the US state – its border regime. As such, it is a key gatekeeper to access information about entering, or staying in, the US. The friendly chatbot projection also glosses over – at least implicitly – both the strong push by corporate actors for designing, testing and deploying AITs in the public sector, and the existence of highly contested AIT applications in immigration and border control in the US from aggressive robot guard dogs and underground fiber optic sensors at the highly policed border with Mexico (Fussell 2022) to the (unsolicited) scanning of social media accounts to enforce "data-driven deportation" of residents without a valid permit (Bedoya 2020).

So what is the policy, political, legal and administrative work actually done by Emma, and other AIT applications in public policy contexts, both visible and hidden? How is the relationship between these diverse technologies, public policy and the state, unfolding, and with what effects: to facilitate or obstruct public policy design and delivery; to improve, disrupt, advance, corrupt the role of the state, government and public sector; to change what states do and how they do it; to transform how states, corporations and citizens relate to each other, where power is located and how it is exercised? This *Handbook* sets out to address these issues across a wide array of geographical, socio-economic, technical, political and policy contexts. It thus contributes distinct public policy and governance expertise to a growing field of critical AI studies (e.g., Lindgren 2023), setting a democratically oriented counterpoint to the high

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pace and latent unpredictability of AI diffusion in the public sector, but also to the concentrated politico-economic interests, techno-solutionist framings and high-flying promises for a future-oriented public sector that underpin it.

Tensions and Controversies in AI and Public Policy in a Nutshell

Characteristic of wider arguments in favour of introducing AITs in the public sector, chatbot Emma was designed to manage the high volume of "customer" queries (reportedly over 14 million calls on immigration issues each year); reduce the workload for (expensive) human agents; and offer targeted help for people searching the webpage. This promise of AI-driven optimization and rationalization aligns well with wider hegemonic visions of what government, public sector and state are for, and how they should be organized.

In the last fifty years, Western-dominated global narratives about how to improve public policy have centred on a perceived need for cost-effectiveness, value for money and improved "customer" (citizen) experience. These originated in agendas for reform that were promoted by management consultancies and academic economists in the Global North (and imposed by international financial institutions such as the International Monetary Fund) under the global brand of "new public management". Reforms varied widely in practice in their extent, form and outcomes. Nonetheless, overall, they challenged conventional ideas of bureaucratic and rule-based efficiency through which states exercised collective rule, and promoted market-based thinking, in which public sector organizations should be managed through performance indicators and budgetary responsibility, while responding to citizens as customers (seminal overview in: Hood 1991).

These reform agendas went hand in hand with the development of digital technologies, from e-Government to mobile technologies, closely aligned with the presumption that the citizen is a consumer: at first, offering information, and eventually, for those with digital resources, enabling citizens to access information, complete forms or apply for licences, anytime, anywhere. At the same time, facilitating the push for "value for money", monitoring procurement contracts and meeting performance indicators, digital technologies facilitated the collection and analysis of budget data points, operational efficiencies and outputs. Privileging the contribution of digital technologies to "better" policymaking led to techniques of "agile policymaking", which originated in digital product development, being adopted as innovative policymaking methods. Also promoted by management consultancies and international organizations such as the OECD, these sit uncomfortably alongside ever more detailed monitoring of performance and activities of operational and "front-line" staff.

In this context, digital technologies have been offered as the necessary technical solution to the political, policy and operational problems of governing much before the advent of AI. They have been fundamental to the refashioning of citizens as consumers and the state as cost-effective service-provider, but also contributed to the retrenchment of public sector capacities and the development of surveillance technologies able to control public sector workers as well as the wider population (Collington 2022; Fourcade and Gordon 2020; Yeung 2023). Over the last decade, public officials started discovering AITs as a way of radically enhancing their already-existing "technological solutionist" agenda with promises of accuracy, speed and automation in delivering citizen-facing services, providing security and through back-office efficiencies (Veale and Brass 2019; see overview in: Paul 2022). The European Commission, for instance, argues that "AI-enabled solutions in the *public sector* can deliver shorter and

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richer feedback loops for all levels of governance, providing an opportunity to speed up, improve the efficiency and effectiveness of service delivery" (European Commission 2018b: 19, highlight added). This appears to offer the possibility of improving the quality and consistency of public services, help fine-tune the design and implementation of policy measures, render interventions more efficient and targeted, and enhance the cost-effectiveness of public procurement.

However, we already know that (as yet) many of these claims are overblown or simply untrue (on the AI functionality fallacy see: Raji et al. 2022). This is the case already at a technical level. Exploring our opening example a bit further, Emma's designers have programmed the possibility of "machine failure" into the chatbot: Emma leads users on to human agents when it cannot find the right information, or when visitors (re-)code Emma's answers as not having addressed their queries (a familiar experience for most of us using these relatively unsophisticated chatbots). This anticipation of inevitable "machine failure" speaks to a wider discussion in computer sciences, the humanities, jurisprudence and the social sciences: how autonomous, intelligent and reliable are AITs really, how should, and can, we control AIT systems, and are technical fixes enough, possible or even desirable in public sector use contexts where people's fundamental rights and democratic values are at stake?

These questions are pertinent, because in many cases the consequences of "failure" of an AIT system are much more severe than with Emma the chatbot (i.e., not finding the correct information on immigration procedures in the US). Indeed, the techno-solutionist promise of AIT in public policy is repeatedly jeopardized by individual national scandals. AI-driven automated decision systems have produced illegal effects, for example, in Australia ("robodebt" scandal), the Netherlands (child benefits), Sweden or Britain (education), the US (staff performance assessment) (Henriques-Gomes 2021; Peeters and Widlak 2023; Satariano 2020) – and several chapters of the *Handbook* dive into such cases in more depth. Indeed, the first fine ever to be paid under the EU's General Data Protection Regulation (GDPR) was in 2019 by a Northern Swedish municipality (20,000 euros) which had trialled facial recognition technology to monitor students' school attendance and to relieve teachers from the time-consuming work of keeping handwritten records (European Data Protection Board 2019). A number of these systems were withdrawn after legal or political constestation over bias, data protection violations or incorrect outputs. Importantly, however, a common threat in their scandalization has been around more fundamental questions of how algorithms, AITs and automated decision systems transform the relationship of state and citizens. These scandals also sit alongside political and civil campaigns opposing AI deployments on the grounds of the wider public good. When two dozen cities and states in the US started banning facial recognition technologies from 2019 onwards, or citizen protest halted a "smart city" proposal in Toronto, Canada, campaigners had not only worried about individual rights (such as data protection or privacy) but also about undemocratic corporate and state control of the public space (Berger 2020; Simonite 2021).

Apprehensions of the potentially detrimental societal consequences of AI design and use also feature in education, care, health care and policing. For example, the UN special rapporteur on disability has raised pointed worries about state applications of AITs and the discriminatory effects on vulnerable populations (United Nations 2022); Human Rights Watch (2019) has documented profound concerns over the potentially life-threatening effects for people in extreme poverty of AI-fuelled identification and fraud detection in welfare programmes across the globe from the Indian Aadhaar project to Michigan's MiDAS; and UNICEF (2021)

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has published a comprehensive description of risks and policy guidelines for AI and children. Allegations in these areas often rest on the conviction that technical fixes just won't do when a technology internalizes and amplifies societal biases and automates patterns of exclusion in public policies and public services (Benjamin 2019; Eubanks 2018; Singh and Jackson 2021).

Techno-solutionist representations of AITs in policymaking also disguise how the design, development and use of these technologies is driven by corporate interests, and that the use of AITs *in* public policy is also structured by the development of public policies *for* AITs. There are numerous reasons why governments are developing public policies to promote the research design development and use of AITs more widely. These include national projects of technological specialization, developing and maintaining high-value economic activity in design and manufacturing, and the protection of key capabilities, especially in defence and security (Bellanova et al. 2022; Calderaro and Blumfelde 2022; Mügge 2023; Paul 2023; Smuha 2021b). Indeed, the design and development of these technologies raise numerous challenges for public policies, both within and between countries/regions, which reach beyond fundamental rights and undemocratic surveillance of citizens: the security and safety of national infrastructures; the global regulation of precarious labour conditions in the production and maintenance of AIT systems from data to minerals (Crawford 2021; Wood et al. 2019); and perhaps most importantly, the contribution of AITs to combatting or worsening the climate crisis: including energy consumption for large data-dependent AITs, biodiversity and environmental degradation from undersea cable-laying and open-cast and artisanal mining of precious metals, and the disposal of components and electronic waste (Dauvergne 2020; Tamburrini 2022; van Wynsberghe 2021). This means that any high-flying positive assumptions as to the 'cost-effectiveness' contributions of AITs – including the evaluation of chatbot Emma's ability to save Federal resources – rely on naïve or wilful blindness to, and wholesale externalization of, the vast environmental, energy and social costs of producing, recycling and disusing AITs.

Three Vantage Points for Critical Inquiry of AI and Public Policy

Overall then, there are significant unresolved (and perhaps unresolvable) tensions in – and controversies about – the relationship of public policy, governance and AITs. In this opening chapter to the *Handbook*, we develop a conceptual-analytical framework for examining "public policy and AI" against this broad backdrop. Our aim is to open discussions and scholarly evaluation beyond narrow technical and positivistic understandings of both AITs and policy. We propose three key vantage points for critical inquiry that are further elaborated and nuanced in the thirty subsequent chapters of this *Handbook*:

- *the politics of conceptualizing AI*, of defining its role and workings in the policy world, including through the normalization of contestable normative precepts for democratic AI governance such as "ethics", "transparency" or "trustworthiness";
- the contingent practices of *AIT design, development, use and regulation* in public policy contexts, understood as highly variable socio-technical assemblages which pull together specific sets of meanings, strategic actions, technological affordances, forms of materialization and their effects;
- the situatedness of AI/policy interactions and their effects *in a highly uneven global political economy of AIT production and regulation*.

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The remainder of this introduction carves out these three conceptual-analytical vantage points, and their respective elements, in more detail and ends with a short presentation of the *Handbook*'s structure and contributions.

VANTAGE POINT 1: CONTESTED EPISTEMOLOGIES OF PUBLIC POLICY AND AI TECHNOLOGIES

How we conceptualize and discuss the interaction of AITs and policies, and how we frame risks, challenges and opportunities of AITs, depends a great deal on our understandings of what AI *is*. And this is where the struggles begin. This is not because it is per se harder to define AI than to define democracy, poverty or sustainability. Rather, it is that each definition relies on, and reproduces, particular assumptions, both ontological (what is AI) and epistemological (how can we know what AI is and does). And these assumptions are situated in specific value systems and political struggles, which can be implicit, explicit or completely hidden, in the way AI is understood, discussed, used and regulated.

Critical social scientists suggest that AI (and "algorithms") are "ambiguous concept(s) which some see as a technology, some as a decision tool, others as an epistemology or as a socio-technical process of ordering" (Barocas et al. 2013: 2). We need to explore that ambiguity first, to explain and critically evaluate the politics of defining AI. Although our intention is not to define AI "once and for all", it is necessary to state this volume treats it primarily as a cluster of *technologies*, rather than as scientific discipline or epistemology. We propose that AI in policymaking can only be understood in its specific material expressions as a particular *AIT*, because AI "always relies on other technologies and is embedded in broader scientific and technological practices and procedures" including both software and hardware (Coeckelbergh 2020: 80). This embedding in other practices includes the conceptualizing of AITs through their material production and functioning, and through the discourse, mythologies and politics of technologies in general. In this section, we illuminate the contested business of conceptualizing AI – of delimiting what AI is and is not – to show how such definition work *matters* for how and why AITs are inserted in policymaking and how they get regulated. Since concepts of AI, and concepts with which AI is regulated, are themselves part of the politics of using AI in the public sector, this is where critical understanding begins.

Techno-functional Conceptualizations of AI

It seems quite straightforward to conceptualize AITs through their functions and technical characteristics; through "a system's ability to interpret external data correctly, to learn from such data, and to use those learnings to achieve specific goals and tasks through flexible adaptation" (Kaplan and Haenlein 2019: 15). Such definitions highlight the technical capacities of AITs in reflexive data processing and adaptation – understood as "learning" – and a view of these capacities serving a specified function or set of functions. This historically might have been through the AIT being programmed to achieve specific goals and tasks, whereas contemporary AITs involve semi-autonomous goal identification and adaptation.

Techno-functional definitions of AI span technical and policy worlds. They reference concepts, components and types in a cluster of diverse technologies, with empirical and methodological descriptors of varying generality. These include algorithm, machine learning, neural

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networks, deep learning, natural language processing, image recognition and automation/automated decision-making (ADM). A simple definition of AI systems in a recent regulation proposal, for example, treats it as "a collection of technologies that combine data, algorithms and computing power" (European Commission 2020: 3). Another recent Communication by the European Commission (2018a: 2) defines AI systems as "systems that display intelligent behaviour by analysing their environment and taking actions – with some degree of autonomy – to achieve specific goals". AI ethicist Marc Coeckelbergh (2020: 64) relates AI to algorithmic calculations, suggesting that "AI can be defined as intelligence displayed or simulated by code (algorithms)".

There is an ambivalence in these "techno-functional" definitions. On the one hand, they short-cut a long-standing debate within AI, in which what it means to "simulate" intelligence, as well as what constitutes "intelligence", is highly contested. Indeed, understandings of AI date to the original coinage of the term "artificial intelligence" in 1956 by John McCarthy (a computer scientist at Stanford), who organized a two-month research series on Artificial Intelligence at Dartmouth College, New Hampshire. The goal was to create a field of research for "building machines able to simulate human intelligence" (Haenlein and Kaplan 2019: 7). The idea of "simulation" speaks directly to Alan Turing's seminal work on "Computing Machinery and Intelligence" on the design of intelligent machines and ways of testing how well they simulate human intelligence. While Turing did not use the term AI yet, for many Turing's famous reflections serve as a litmus test today: "if a human is interacting with another human and a machine and unable to distinguish the machine from the human, then the machine is said to be intelligent" (Haenlein and Kaplan 2019: 6).

On the other hand – and somewhat contradictory to the anthromorphization of capacities and agency ascribed to AITs – policymakers, regulators and more positivist scholars define AI primarily in terms of its technological characteristics. The common use of analogies to human experiences (learning, adaptation, intelligence) generates an anthropomorphic ontology for AI. The machine (or technological system) is presented as possessing qualities that are external, but similar, to humans. The implication is that machines are inserted as ontologically distinct entities into human activities, institutions and decision-making processes. This has become even more pertinent since the introduction of Generative Pre-trained Transformers (GPT) language models qualitatively shifted AI capabilities and scale, with a potential to power a wider set of applications across the policy world in open or hidden ways and to fathom even stronger notions of AI agency (one of our contributions maps the burgeoning but problematic uses of ChatGPT in judicial decision-making). For this *Handbook*'s purposes, these definitions signal that the AITs have independent agency in the process of making and delivering policy (hence also the fears of bureaucrats losing discretion to machines in parts of the literature).

More critical AI scholars – including computer scientists – have contested the usefulness of the Turing test as well as the implied notions of "intelligence" in the context of AI (Broussard 2018; Chrisley and Begeer 2000; Christian 2020). They warn us that treating a technology meant to simulate human capacities (speech, visualization, image recognition, etc.) based on stochastics and mathematical probabilities as "intelligent" obscures how the technologies actually function. In doing so, it hides not only their inherent technical limits, but also how they interact with socio-political processes. This makes it more difficult to clearly identify, or challenge, the political and economic agendas of boosting their sales and use despite such problems (see foundational paper on "the dangers of stochastic parrots": Bender et al. 2021).

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What holds the techno-functional definitions coherent – and permits the slippage between narrow techno-functionalist definitions into anthromorphizing the machine – is the underlying conceptualization of independence of the machine from the conditions of its production, the unfolding of its functioning or the effects of its outputs. As our contributors discuss, these conceptualizations lead to blind spots in understanding the political role of AITs in public policy, their strong politico-economic underpinnings, and the possibility and practice of their regulation.

Socio-technical Conceptualizations of AI

In contrast to techno-functional definitions, many social sciences and humanities scholars refute the ontological independence of algorithms, AI and AITs, arguing that, instead, they only exist and can be understood as existing in wider social relations (e.g., Barocas et al. 2013; Coeckelbergh 2020; Katzenbach and Ulbricht 2019; Ziewitz 2016). Furthermore, from a political lens, to consider AI as comprising cognition, reasoning and decision-making that is independent from social interactions is simply utterly misleading. AITs are socio-technical assemblages which – due to their technical functionalities – may represent a specific cognitive and material form of collating and processing data; but this is necessarily always embedded in, and co-produced with, social processes (for the wider argument about socio-technical assemblages, see: Åm 2015; Jasanoff and Kim 2013). Critical sociologists demand that our analyses look beyond the figure of the algorithm in any essentializing way and attend, instead, to "how multiple spaces for agency, opacity, and power open and close in different parts of algorithmic assemblages. The crux of the matter is that actors experience different degrees of agency and opacity in different parts of any algorithmic assemblage" (Lee 2021: 65). As Jasanoff (2016) puts it succinctly, "technological objects are thoroughly enmeshed in society, as integral components of social order". And as such, AITs are "artifacts [that] have politics" (Winner 1980), directly and profoundly relevant to public policy and governance. AITs are not only facilitated by political action or used in political institutions to achieve particular goals. They are also materializations of, and themselves facilitate, relations of inequality, domination, possession and extraction through their design, development, use and disposal (Benjamin 2019; Couldry and Mejias 2019; 2021; Gray 2023; Kwet 2019).

Contributions in the *Handbook* therefore explore the role of AITs, algorithms and data-driven decision-making as part of socio-political ordering processes, including "the economic, cultural, and political contexts that both shape the design of algorithms as well as accommodate their operation" (Katzenbach and Ulbricht 2019: 3; cf., Burrell and Fourcade 2021). For example, an already large literature on critical data studies highlights that AITs (and algorithmic calculations more generally) rely on socially constructed and *always already* pre-interpreted data. Data is never "raw" but emerges from highly contingent ways of categorizing and labelling the social world (Gitelman 2013), and from the unequal distribution of "symbolic power" (the power to name and label things in dominant ways) (Bourdieu 1991). Data and data-based technologies for sorting and classifying people – key uses of AITs – have a "looping effect" (Bowker and Starr 2000; Hacking 1995; Law and Ruppert 2016), coming to shape people and their understandings of themselves over time, and in turn being reshaped by the actions and reactions of those people.

The contrast between techno-functionalist views of AITs (as neutral and rational decision-aids) vs. socio-technical understanding (as integrated in socially constructed gov-

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ernance tools in socio-political ordering processes) echoes longer-standing debates about quantification, analytical models and technical expertise in policy studies. More positivist accounts highlight the role of technical tools, visualizations, numerical analysis and predictive models as neutral informants of rational problem-solving in the policy world (e.g., Majone 2010; Posner 2001; Sunstein 2002), while more constructivist analyses of public policy and governance highlight the inescapable connectedness of these tools, analyses and models to processes (and contestations) of socio-political ordering (e.g., Desrosières 2002; Paul 2021; Porter 1995; Stone 2020).

This chasm also speaks to a wider controversy in policy studies about the epistemological features of policy itself, which predates the arrival of "analytical turns", let alone AITs, in public decision-making. Indeed, Harold D. Lasswell's influential concept of "policy science" ascribes a *neutral* and *rationalizing* role to policy analysts whose expertise informs the search for optimal solutions to politically predefined problems, assisting "the realization of the value goals" (Lasswell 1970: 5). Even though Lasswell acknowledged that the use of policy knowledge was always "contextual" because it will move through differently structured polities and political interactions, in his work, policy expertise itself retains its realist epistemology. As one of us argues elsewhere, many policy scholars conceptualize the more recent rediscovery of analytical tools in policymaking – from risk analysis to cost–benefit analysis – as a Lasswellian move of rationalizing policymaking (Paul 2021). This view clashes with *critical policy studies* which see policymaking as a much more complex, ambiguous and contested process of vesting meanings, normative judgements and selective representations of social reality in a "policy" at the discursive, epistemological, legal as well as practice level (Carmel 2019a; Fischer et al. 2015; Stone 2012; Yanow and Schwartz-Shea 2006). Scholars in this camp treat knowledge claims and their materialization in specific tools, models and institutions as an intrinsic part of the politics of policymaking. This is perhaps doubly important to acknowledge when considering AITs, as they are so reliant on datafication for their technical functioning all while their processes of modelling reduce the expression of contestation, ambiguity and complexity to categorizations and labelling of huge numbers of seemingly unambiguous and neutral data points. It is, in this case, not just that the AI-powered tools themselves present the idea of a rational, truthful and objective analysis (e.g., the identification of a medical risk score for prioritizing patient treatment), but that their entire functioning rests on assumptions that the data points they processed for such analysis are expressions of reality onto which a notionally rational analysis can simply be imposed.

The Role of Law and Regulation in Shaping AI Public Policy Interactions

This wider critique, or orientation, also brings us to consider the role of public policy and law that we can see in burgeoning efforts to regulate the effects of AITs across all scales and diverse contexts and applications. To assess this development, we also adopt a socio-political account of law and technology. Law is constitutive of social relations and political economy and plays an important role in spurring, shaping, and … technological developments. Legal definitions of AI – though primarily techno-functionalist – do not form apolitically as merely a neutral description of a particular technology or activity. Choices about how to define certain terms in law – what to include or exclude, how to include or exclude – are necessarily *political* choices. They are the product of calculations, rationalizations, problematizations and are tied to political and ideological views and assumptions as well as policy considerations and goals.

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Indeed, though a common view is that law "falls behind" technology or that law must simply *respond* to how technologies are being used, the relationship between law and technology is a cyclical, symbiotic, *coevolutionary* one. The work going into defining the issues to be regulated by law – AI, "bias", "privacy", "transparency" and so on in our case – generates the framework for which issues in which cases will be ruled by certain legal principles, rules and sanctions.

While this is highly political – and often contested – work in the first instance, legal definitions tend to become sticky and neutralized over time in ways that are no longer visible or debated. The law involves selecting specific permitted relations, actions and processes in society from a much wider set of possibilities. It normalizes and generalizes particular political choices by inscribing them into law and regulation (cf. Paul 2022). In doing so, it also organizes, or makes less visible, the struggle and contestation over those choices. The by now abundant insights from ethical and socio-legal research on AITs' effects on fundamental human rights, rule of law procedures in public decision contexts and other normative baselines for democratic governance (e.g., Ananny 2016; Cobbe and Singh 2020; Coeckelbergh 2020; Dubber et al. 2020; Pasquale 2015; Smuha 2021a; Yeung 2019) highlight that such technology-aided categorizations have profound effects on people's lives. Critical analysis of AITs and public policy must discuss the discriminatory and marginalizing effects of AIT uses. This is especially relevant where official policy or regulatory rhetoric silences or marginalizes them as matters of *individual* rights, even where discriminatory patterns are in fact much more structural in character and would require a wider focus on social justice and equity (cf. on the disregarded "collective dimension" of individual data protection regulation: Mantelero 2016).

A Heuristic for Exploring Public Policy and AI Technologies

For this *Handbook* to unpack the contested ontology of AI, its overlaps with equally contested policy epistemologies, and the relationship between using AITs in the public realm and regulating them we propose a novel heuristic (Table 1.1). This provides a first rough compass for critically situating existing research and policy proposals in a two-dimensional conceptual space of interactions between public policy and AITs. The heuristic provides us with a device to systematically map state-of-the-art research on AI and public policy, for analysing empirical phenomena and policy debates, and for launching informed critique about blind spots of existing research and policy debates. It enables contributors in this *Handbook* – and hopefully other researchers and critical policy analysts after us – to expose what remains hidden from view or is deliberately marginalized through techno-functional interpretations of AI's ontology in policy and regulation.

VANTAGE POINT 2: MEANING-MAKING, POLITICAL ACTION AND MATERIALITY IN AI POLICY PRACTICES

Next to mapping the relative weight of techno-functional AI ontologies in our field of studies and in the policy world, the *Handbook* also aims to examine the implications of socio-technical understandings of how AIT and public policy intersect. In particular, we suggest that a "practice"-oriented approach to public policy and governance can illuminate how actors justify and enact the role of AITs in public policy in different settings "on the ground". We do this with

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Table 1.1 Heuristic for exploring public policy and artificial intelligence technologies

		Perceived ontology of AIT/policy	
		Techno-functional	*Socio-technical/political*
Relationship between AIT and policy	*AIT as tool in policymaking*	AITs as techno-solutionist decision aids (implicit: policymaking as rational problem-solving)	AITs as governance tools in socio-political ordering processes (implicit: policymaking as political)
	AIT as regulatory issue	Narrow AIT regulation, (implicit: e.g., risk-based regulation as rational problem-solving)	Broad AIT regulation, (implicit: regulation as struggle over wider power relations, capitalist accumulation patterns etc.)

a view to exploring how and why different contexts generate different relationships of AIT and public policy through a three-fold focus:

- actors' meaning-making about AI and its role on policy as constituted in a specific context;
- policymakers' strategic but unpredictable interactions with specific technologies;
- forms of materialization which normalize specific socio-technical imaginaries and actions and thus structure future policy and practice regarding AITs.

Public Policy as Practice

A practice-oriented approach enables us to access three important aspects of policymaking: meaning-making, action and materiality. Critical policy analysts understand policymaking as mediated by "communicative practices" which are mobilized or silenced in specific empirical settings in contingent ways (Fischer et al. 2015: 5; Dubois 2009). This orientation requires exploration of AIT/policy interactions beyond the level of formal policy or regulatory programmes on paper. We need to examine in context, how high-flying promises of accurate, cost-effective and speedy decision-making with AITs come into being, are realized, or not, how they are contested, mediated or appropriated by all involved policy actors, each facing diverse constraints and drivers to action. "Socio-technical imaginaries" shape how individuals, institutions and cultures consider the range of possibilities, threats and proper uses of technologies, and these are themselves saturated in relations of power to define what technologies are and who are their owners, users or objects in use cases (Jasanoff and Kim 2013; also Lewis et al. 2020). But while the politics of "talking AI into being" through specific policy frames has already received increasing attention in research (cf. af Malmborg 2023; Bareis and Katzenbach 2022; Paul 2023), it is in specific use cases on the ground that preliminary normative calls for (or against) using and regulating AITs in policymaking and public service provision, political struggles over these uses, as well as patterns of domination in whose judgement matters, can be revealed.

But practices are not just about "meaning-making". As political actors engage in policymaking, they act: forge alliances, distribute resources, give privileged (or deny) access to other actors. In acting, actors' contingently synthesize ethics, knowledge, normative understandings, interests, strategies and goals for action: whether collectively or as individuals (Shore et al. 2011; Shore and Wright 1997; Wagenaar and Cook 2003). Practices have a concretization function for public policy: they bring policy "into being" on the ground beyond formal decla-

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rations of policy goals. But in that grounding, practices also expose tensions between conflicting goals, norms, interests, understandings and so on; and they require actors on the ground to work with, around or against any ambivalences, ambiguities or outright contradictions. Contributions in the *Handbook*, especially in Part IV, illustrate what such a practice lens can reveal about the AI/public policy complex.

In our particular case of public policy's technology orientation, practices involve interactions between human and non-human actors on the ground which are strategic from the side of policymakers but have unpredictable effects. As Henk Wagenaar (2012) stresses, in his definition of what makes policy "actionable", "*practice* is a move or thrust into an only partly known and knowable world. By acting, by intervening, we extend our intentions and understandings into this indeterminate world without being able to predict how its agency will effectuate itself and impact us".

This view on policy, as a practice where policy actors enact their intentions and meanings but still throw themselves into the partly unknown, also concurs with more recent discussions of "technological agency" as an "*iterative process*, … in which humans construct technologies that impact their users and designers through unforeseen uses and effects" (Campbell-Verduyn and Hütten 2023). Despite its unpredictability then, the contingent and mutually constitutive iteration between policy goals, interests, intentions and meanings vs. technological affordances and their effects in AI-related policy practices is not neutral, accidental or chaotic. We suggest that policymakers act purposefully and strategically in relation to AI – and their actions are structured by inequalities in power, knowledge and resource, and by the meanings they vest in their specific action (Carmel 2019b; from the perspective of Sum and Jessop's cultural political economy: Paul 2023).

From the combination of meaning-making, structuration and action, then we come to a third important aspect of practices, that they "involve the *materialization of* meanings, knowledge and ideas in concrete objects, institutions, and the relations between [them]" (Carmel 2019b: 37, emphasis added). This focus on policy as a continuous, complex and contradictory process of materialization, enables us to explore how the material objects and infrastructures; the component parts of AITs, intersect with the socio-technical imaginaries of what we think such technologies might do or be. In our case, this materialization might be seen in the use, location and networking of data centres; the design of public procurement platforms; the age and structure of existing computer networks; the institutional organization of budget-holders and reporting; datasheets and audit protocols; or the design of user interfaces used by decision-makers. Each of these component parts and processes intersects with what different policy actors think about what is possible or right to do when using AITs – from the national strategy team, to data engineers working on the technology, to social workers and border guards trying to use (or seeking to evade) it.

At the same time, our attention to the materiality of policy's interactions with technologies does not ontologize the material world as external to social production processes. A recent upsurge of critical infrastructure studies (Bellanova and de Goede 2020; Bernards and Campbell-Verduyn 2019; The Critical Infrastructure Collective 2022) shows how the material underpinnings which lend infrastructures – roads, the internet, currencies or the cloud – their sense of durability and material existence, have been socially created and are being constantly reinterpreted and struggled over in everyday practices. Our conceptualization of policy as practice thus also exposes the politics of creating, maintaining and normalizing material structures as a crucial part of existing AI/policy interactions.

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Co-production of Technologies and Social Order

Our view on AI policy practice speaks to the older science and technology (STS) assumption that the development, uses and regulation of technology cannot be explored in the abstract: we agree that these processes are inexorably linked and *situated* in specific socio-technical contexts in which technology is *co-produced* with societal knowledge, institutions and power (Jasanoff 2016; Jasanoff and Kim 2013). Rather than focusing on AIT uses in narrow cases, contributions in this *Handbook* provide a contextualized understanding of socio-technical systems as "contingent on social, political, and economic forces and can take different shapes" (on "algorithmic governance": Katzenbach and Ulbricht 2019: 7). While proponents of "trust-worthy AI" or "proportionate" regulation of ADM systems acknowledge that context matters for assessing AI-related risks and benefits (Chatila et al. 2021; Krafft et al. 2022), we are yet to consolidate critical policy-analytical insights on how the public sector *actually adopts and incorporates* AITs. In particular: how are these technologies being mapped onto or disrupt existing institutional practices, how are such diffusion processes shaped by sector-specific norms, and how do actors in different contexts enact and contest abstract regulatory safeguards such as "accountability", or protection against "bias" on the ground?

A comprehensive review of public sector uses of AI in the US includes the 142 most significant Federal Agencies and Departments (Engstrom et al. 2020). This concludes that while 45 per cent of these have experimented with machine learning, there is a grave lack of technical capacity and knowledge within the administration that defies reflexive use and careful inspection of AI-based decision recommendations. And there is more than one in which appropriation of technologies can unfold. For the British police (Sandhu and Fussey 2021: 79) highlight "a range of fissures" occurring as police officers find the smart technologies they are expected to use flawed, and resist AI's intrusion of their discretion. This also reflects studies in the Canadian border force, although Côté-Boucher (2020) found variation in how officers endorsed or resisted the AIT recommendations. In a rather different context of technological appropriation, Koulish and Evans (2021) show that in the US border force, officers overrode an original automated decision recommendation system so that it better conformed to their norms/preferences. Several system redesigns tried to better match the AI system to the officers' decisions, eventually resulting in the automation of their discriminatory decision-making. Another study on police uses in the Netherlands shows that officers only adopted AI-recommended decisions when they matched their own intuition (Selten et al. 2023). These contrasting cases imply diverse relationships between different policy actors, institutions, norms, expectations and how the technologies function over time, in which we have yet to have a systematic explanation of these variations. A practices perspective that examines meanings, action and materiality is critical to understanding individual cases and to explaining varieties of AIT/public policy relations in different contexts.

Importantly, these cases also bely references to the generic "human" of classic AI ethics literature, and highlight specific contributions of comparative law and regulation in examining what "human control" might mean, and how it is enacted in highly variable and contingent ways, across policy sectors and jurisdictions (Elish 2019). These discussions also highlight how institutional norms, material limits of technologies and our ideas about what technology is, profoundly shape the regulatory meaning of "control", "accountability" or "liability" (Firlej and Taeihagh 2021).

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We propose that empirical analyses of policy and regulatory practice must address the highly contingent sociotechnological articulation of AITs and human actions in policy, and explore the intricacies of complexly interwoven meaning- and decision-making processes beyond the widespread man/machine dichotomy. In that context, the widespread parlance in AIT regulation of keeping "the human in the loop" (e.g., Firlej and Taeihagh 2021) is utterly counterproductive. It assumes a neat distinction of human/machine decisions in AI-powered policymaking, with humans ready to interfere when – and just when – the machine fails. Such dichotomy is not only *conceptually flawed* (useful discussion of such flaws also in: Jones 2015); it is also *democratically problematic* in public sector applications, for at least three reasons:

- focus on the AIT as a machine actor releases publicly accountable actors from liability for choices on procuring and designing an AI, selecting data, initial coding and programming, or how to incorporate the AI into hardware and how to use it in decision-making; choices which co-shape how AI works in practice and how it affects citizens;
- the "human-in-the-loop" model also turns policy experts and bureaucrats into "moral crumple zones" who soak up liability for machine failure but have little discretion for professional, moral or empathy-based judgement otherwise, and are being increasingly de-qualified in taking decisions in the first place (Elish 2019; cf. Yeung 2019). There is a risk that humans are degraded to mere correctors of machine decisions but cannot and ought not improvise to deal with contingency (Green 2022);
- while actual oversight of systems is therefore currently difficult or impossible to secure, claiming to have this oversight also releases the institution procuring and using such systems from responsibility (Green 2022). Beyond the individual system user who bears a responsibility they cannot exercise, then, the wider accountability of public bodies for any individual or collective harms from the functioning of these systems is also displaced by this "human-in-the-loop" framing.

In conclusion, the empirical exploration of AI/policy interactions necessitates moving beyond dichotomous understandings of human–machine autonomy, decisions, cognition or knowledge. Practice-oriented policy research on AITs needs to tease out the politics of perceiving, enacting and contesting the fluid boundaries and interactions between humans and machine – and the overriding role of encoded norms, values and inequities – in concrete cases. Our socio-technical examination of AI's ontology in policymaking (the first vantage point for critical inquiry) then aligns with STS-inspired examinations of technology *affordances* in policymaking not as neutral tools at the disposal of the rational human actors pursuing their policy goals, but as integral but also rather unruly parts of the social production of order and power.

VANTAGE POINT 3: (GLOBAL) POLITICAL ECONOMIES OF TECHNOLOGY, LABOUR AND NATURE

A focus on meanings and practices, however, can still not fully expose the "political" in the AI/public policy complex. From our *critical* perspective, we need to situate these assemblages in the wider structures of AI production, development and use. To us, this particularly concerns the ways in which AITs and their production depend on and reproduce global, but also more local, power inequalities and injustices, and the ways in which public policy measures in the

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AI domain contribute to specific production patterns and their unequal effects (also see de Freitas Boullosa et al. 2023). Such critique falls into three categories:

- the socio-economic injustices of informational capitalism and its inherent data production processes;
- the environmental injustices created by AIT systems' hardware and energy demands; and
- the larger scale corporate transformation of the state.

Socio-economic Injustices in AI's Production Processes

In efforts to expose the socio-economic injustices of specific AI-fuelled governance projects, policy scholars have already taken pains to specify generic concerns over bias and equity, and contextualized their analyses of distinct AIT use cases in wider societal structures. For example, analysts of "smart" border control policies highlight the risk of state authorities violating migrants' fundamental rights (such as non-discrimination or fair treatment) without noticing these violations, being able to account for their occurrence or correcting them in due course (Amoore and Raley 2017; Leese 2014; Molnar and Gill 2018). Scholars of predictive policing point to similar dynamics and discuss the self-fulfilling prophecies of algorithmic racial profiling, targeted interventions in "Black" neighbourhoods, "finding" crime and feeding this "information" back into an algorithm that has been trained with data that is *always already* racially biased (Ferguson 2017; Kaufmann et al. 2019). Social policy analysts equally worry about automated forms of discrimination which could deepen structural inequalities in access to social benefits and services (Dencik and Kaun 2020; Eubanks 2018; Redden et al. 2020).

Such focus on the societal injustices stemming from, and/or perpetuated through, uses of AIT systems in the public sector remains crucial also for contributions in this *Handbook*. However, if we more closely consider AI production at the level of data, additional socio-economic injustices come into view. We suggest that the political economy of AIT, and its interactions with public policy, emerges from the wider systems and structures of informational capitalism, in which data – its collation, curation, labelling and "refining" (Cohen 2018), parsing, aggregation, reconstitution and use – is central. For Cohen, informational capitalism depends on the imagining and constitution of what she calls a "biopolitical public domain": a "construct that enables both the marshalling of informational resources and the exercise of biopower over subject populations" (Cohen 2018: 215). Technological systems (cookies), legal precepts (consent, privacy, intellectual property) and the interest in selling products (targeting potential buyers) combine to facilitate a drive to appropriate the resources of this public domain. Platform economy companies such as Google, Amazon, Meta or Apple drive the mass generation, extraction and processing of user data at the heart of their business models. It is not data per se that is the asset to be traded and capitalized, as Zuboff (2019) argues, but the "behavioural surplus" generated through the algorithmic structuration of data into patterns that enable customers – such as advertisement companies – to predict individual behaviour and preferences and micro-target interventions. Monopoly endeavours of these companies as part of the business model of "data rentiership" (Birch et al. 2020; cf. Petit 2020).

Importantly, this data-generation is also produced and curated by workers. Thousands of data-workers, employed through less well-known firms and their chains of sub-contractors, spend all day every day coding and (re-) labelling data on which AITs depend. These are often

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workers who sit in unexpected relations to the forces and flows that mark global capitalism. They are often well educated but also marginalized and low-paid, based in the Global South but in key, digitally connected and well-serviced hotspots (e.g., migrant workers and women in Venezuela, Kenya, Philippines and Bulgaria (Posada 2022; Miceli et al. 2020), as well as prisoners in countries across the globe, from Finland to China (Hao 2019)).

"Generative AI" is a particularly pertinent example of these systems of production (see Hao and Seetharanam 2023). The proprietary "ChatGPT" produced by the innocuous-sounding company "OpenAI" (in reality, Microsoft) was released in early versions for use without a fee (the so-called "freemium" model), so suggesting the idea of something automatic and cost-free, just sitting there, as an interface on the web, for fun or as a convenience. Yet, the production of this tool not only extracted and used the creative work of humans without attribution or payment (e.g., works of literature); and the freely donated and enthusiastic interactions of millions of users to adapt and improve the tool for later sale as proprietary product. It also required millions of US dollars to produce, process and label the data that underpins it. In that context, it is not sufficient to discuss racial and other forms of "bias" and discrimination in AIT design and practice (Benjamin 2019; Eubanks 2018); but there is wider political economy of data-dependent AITs in which discriminatory effects are somewhat cynically produced by a highly unequal global labour infrastructure which supports the technological wonders of AI (Casili and Posada 2019; Irani 2015).

The intersection of AITs and public policy thus raises long-standing questions of labour rights and social justice, corporate power, and the role of the public policy in regulating these, not only over the use of AI to govern work and workers (e.g., Amazon in the US), but also in the use of gig forms of work to produce AITs in the first place. This has led to legal contestation (against Meta in Kenya, for example); calls for data justice and new forms of regulation; and to calls for participatory data work. Yet these approaches still leave the solution "in place" – where this data work is undertaken. The real challenge for the relationship between AI and public policy is to make more visible, and to act on, the connection between the production of AITs, and their consumption (for a comprehensive and insightful review, see Dencik et al. 2022).

Socio-ecological Injustices of AI Production

Secondly, the structural injustices of this political economy are intensified by the socio-ecological injustices of AI production – again not usually central to existing regulation and policy studies in the AI domain. As Dauvergne (2022) argues, while there are efficiency gains from the use of AI, in energy management and use of renewables, in transport, for geological surveys and for supply chain management (pp. 702–705), but these savings tend to "accelerate extraction" (p. 705), especially at the point of system structure (e.g., hardware and infrastructure production). The marginal efficiency gains are used to extract profit and intensify production; the efficiency gains of AI at the micro-level outweigh the costs at macro-level. This is especially so since such costs are usually "externalized"; i.e., neither borne by corporate producers nor key users of AIT systems (see below and cf. discussion of the "responsibility gap" for environmental costs of AI diffusion by Lucivero, Chapter 12 in this volume).

Transnational corporations are, at the point of extraction, doing "everything in their power to spur inefficient consumption of natural resources, reengineering cultures toward hyper-consumerism, and designing products for disposability, rapid obsolescence, and fast

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turnover" (Dauvergne 2022: 711). Such hyper-consumerism of AIT products and services is already taking hold in the public sector, too – not least because dominant techno-solutionist narratives of AI boosting cost-effectiveness and rational problem-solving in public service delivery may unfold isomorphic pressure among policymakers and public sector organizations across the globe. The emulation of AIT practices based on states' wish to seem future-oriented is likely to continue without due attention to whether specific applications' purposes and public value are worth their carbon emissions, energy consumption levels and labour costs.

The intensification of environmental degradation produced by mining and petro-chemical processing for plastics required for AIT production is evident in places on every populated continent. And, as Crawford (2021) argues, such intensification is integral to what AITs are, what they promise and how they work. The toxic consequences of "cheap" global production of the hardware of AITs are borne very often by rural communities faced with unsafe, unprotected working conditions and the destruction of landscapes and ecosystems. And they are a feature not only of production, but also AIT disposal, as the global tonnage of e-waste is estimated at 61 million tonnes, predicted to rise to 74 million tonnes by 2030: and less than 20 per cent estimated to be recycled (Statista 2023), with the deleterious ecological effects concentrated in impoverished communities.

Wilful blindness to these costs, and the related injustices, intensifies colonial forms of domination and oppression in the global political economy (Bhambra 2021). Extraction costs in areas settled by formerly colonized and indigenous populations (including the so-called "Global North") are glossed over, disguised or notionally ameliorated. At the same time, technological "benefits" and economic profits are extracted and secured by a relatively small range of companies – many directly colonial or settler-colonial in origin – and their host states in the Global North.[1] All told, this structural condition of AIT production raises profound questions about the "high-tech future(s)" they promise, and whether they are yet – or even can be – compatible with climate goals. For our contributors, such questions require critically discussing whether and how policymakers and regulators address global socio-ecological injustices related to AIT production and how any failure to do so jeopardizes their climate action, (formal) commitment to sustainable development and decolonial agendas (Carmel and Paul 2022).

Transformations of States in the Global Political Economy of AI

Thirdly, and drawing on the previous two points, the state is positioned ambivalently in relation to AI and its production. On the one hand, corporate logics in informational capitalism seem to change how states think about and do policy and governance. With pushes to "mint" data and governing what is measurable, states become an "open data portal" also for private firms and a "provider of digital services", and which thereby "opens itself up to competition from private alternatives that may command equal or greater legitimacy on these terms" (Fourcade and Gordon 2020: 95). Several commentators warn that the diffusion of AI-driven systems will bring a corporate and neoliberal remaking of states, unless data and models are brought into collective public ownership and/or created in more participatory ways (Boyer 2022; Dencik et al. 2019; Pasquale 2016; Pistor 2020) and unless technologies are used for the common good rather than for punishment and control (Bondi et al. 2021; Floridi et al. 2021; Rider 2021).

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Pessimistic accounts even suggest that the digital age will eventually obliterate territorial forms of state organization as a provider for public services and welfare altogether, as informational capitalism enables corporate "data controllers" both to monetarize data/patterns *and* to increasingly determine access to private and public services (Pistor 2020). In addition, the pursuit of an imagined "disruptive innovation" and interstate or interregional competition might lead regulators like the EU to eagerly "[privilege] economic growth over individual rights" in their regulatory proposals (Padden and Öjehag-Pettersson 2021) that further cement the corporate appropriation and valuation of public data. From this then emerges competition among the largest corporations to shape and determine the role of international standards and global regulatory measures in AI to help generate or consolidate global market power, thus reifying and intensifying existing economic and political hierarchies.

On the other hand, claims to state losses of control and capture by tech-corporate power are rather reductionist, and might be unhelpful for thinking through states' complex relationship with data, surveillance and rule in specific applications of AITs (Işın and Ruppert 2019; Tréguer 2019). For example, critical political economy accounts suggest that data appropriation strategies and the push to widely deploy AITs in the public sector might not be a one-sided process of corporate imperialism. Instead, these processes are proactively supported by "AI competition states" (Paul, Chapter 20 in this volume) which think of investment and market enhancing regulation as tools for boosting their national/regional economies' competitiveness in the "race to AI" (cf. Smuha 2021b). As we discuss elsewhere (Paul 2023), the regulatory fabrication of "trustworthy" and "human-centric" AI in the EU's proposed AI regulation can be understood as a tool for "regulating into being" a trademark of ethically superior AI meant to boost the EU's AI competitiveness against the US and China.

AITs did not emerge from a uniquely creative, risk-taking or innovative culture, expressed in the idea of "tech-bros" of Silicon Valley. Computing technologies, including AI, developed from an organized and well-funded "military-industrial-compute complex", to paraphrase C. Wright Mills. Huge-scale public funding of security and defence (in the US notably through DARPA), as well as public university funding, subsidized major technological advancements that underpin contemporary uses of AITs, and continue to provide a steady supply of highly qualified researchers to work (e.g., for Canada: Brandusescu 2021). And local computer networks built around pioneering public schools and libraries were later incorporated by major telecommunication companies for the digital infrastructure that later became the internet (Rankin 2018). Importantly, however, this entanglement of state/public and corporate/private knowledge production is now intensified in the reverse process of appropriating AI research through technology company funding of projects, research programmes, Chairs, university centres and sponsorship of discipline research events (Young et al. 2022). This is critical in shaping how select AI "problems" are prioritized, investigated or addressed, while being liberated from the requirements to contribute to the wider domain of public goods through excessively low and arbitraged taxation. These developments are exacerbated by the material demands of AIT development. They also privilege large technology companies due to the huge resource requirements (energy, computer power, data volumes), in which universities in the Global South, and increasingly in the Global North, are not able to access, without the cash and infrastructure offered by the largest global corporations (see Strubell et al. 2020).

Last, not least, the emerging state-corporate complexes in how AITs are being produced, fuelled with data and research, and used also risk perpetuating as well as creating new forms of colonial domination, extraction and exploitation. The decolonial turn in data and technology

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studies highlights, for example, how processes of data extraction, value generation, the design of applications and the use of AITs in Global South contexts privileges the interests of a few states and corporations in the Global North – increasingly including China – while severely curbing the room for locally owned and independent policy and socio-economic development (Birhane 2020; Couldry and Mejias 2021; Gray 2023; Kwet 2019; Mann 2018). Several of our contributors draw attention to such global power asymmetries by discussing the concept of colonialism explicitly in their work.

ORGANIZATION OF THE HANDBOOK

Keeping with the objective of unpacking the conflictual and ambivalent relationship between AI and public policy, beyond techno-solutionist claims and positivist methodologies, the *Handbook* is structured in four parts which reflect our vantage points for critical inquiry.

Part I, titled "AI and Public Policy: Challenges to Key Concepts", addresses the conceptual foundations for capturing the relationship between AI and policy. How does AI challenge key notions on public administration and policymaking such as power, bureaucratic decision-making or discretion? How do we need to extend these concepts to account for the relationship beyond simplistic man/machine dichotomies and without depoliticizing public policy? How can critical social science contribute to grounding such conceptual discussions? The part starts with a systematic literature review by Andreas Öjehag-Pettersson, Vanja Carlsson and Malin Rönnblom of concepts of "politics", highlighting a surprising lack of engagement with the political nature of AI in social science research. Lena Ulbricht's chapter brings the political centre stage by discussing different concepts of power in AI and public policy, while Roy L. Heidelberg shows that AI in the public sector may be the fulfilment of an ultimately rational bureaucracy which even Herbert Simon and Max Weber might not have believed possible, and which might already have spiralled itself beyond democratic control. The chapter by Frans af Malmborg and Jarle Trondal reflects upon how AITs transform or clash with existing logics of public sector organizations. The part closes with Peter André Busch and Helle Zinner Henriksen examining how AITs reconfigure discretion in street-level bureaucracy and why such transformation is problematic for democracies and the rule of law.

Part II is concerned with "AI and the Politics of Governance: Deconstructing Normative Precepts". Chapters in this part unpack the political work done by guiding principles for AI regulation and governance in different contexts. Authors critically review the following debates in research and policy in alphabetical order: accountability (Jennifer Cobbe and Jatinder Singh), bias (Sun-ha Hong), ethics (Malin Rönnblom, Vanja Carlsson and Michaela Padden), explainability and interpretability (David M. Berry), interoperability (Matthias Leese), sustainability and "digital pollution" (Federica Lucivero), transparency (Ville Aula and Tero Erkkilä) and trust/trustworthiness (Rory Gillis, Johann Laux and Brent Mittelstadt). Taken together, these chapters map the history of related debates and regulatory foci, identify key shifts and struggles over how researchers and policymakers think about these norms, and critically examine the implications of what remains out of scope in research and public discourse on, and the regulation of, AITs.

Part III examines "AI and the Political Economy of Public Policy and Regulation", thus setting the *Handbook*'s contributions in the wider context of AIT production and its implications for global justice. The chapter by Catriona Gray reviews decolonial critiques of

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AI and discusses implications for how we study and develop policy and regulation. Sally Brooks takes on the detrimental socio-economic and political effects of deploying AITs and other data-driven systems, including increases in corporate power in the Global North, in development policy contexts. Adekemi Omotubora and Subhajit Basu discuss colonial path dependence and copycat dynamics in AI legislation and governance frameworks in the Global South and highlight how these marginalize local interpretations of AI risks and harm. Cary Coglianese's chapter examines the two-fold implications of AI for public procurement: challenging how AI systems may be used in public procurement, and the extent to which public procurement may have potential as de facto regulation of AI systems in public decision-making. In his discussion of regulatory interdependence, Daniel Mügge systematizes discussions of global AI governance by reflecting critically upon the extent to which different national jurisdictions can control uses and effects of AITs. Finally, Regine Paul's chapter develops the concept of "AI competition states" to explore states' strategic role in regulating AITs in line with visions of their future techno-economic competitiveness.

Part IV turns to critical examinations of "AI and Public Policy on the Ground: Practices and Contestations". We have curated an insightful collection of chapters across ten selected policy domains to offer readers intimate insights into the wide and growing range of different technologies used, the various public purposes and goals inscribed into deploying AI systems, diversity of practices within the same policy domain across time and space, and dominant struggles over AIT uses in different policy domains. The part includes both widely discussed AIT applications, such as in predictive policing or welfare fraud detection, and a range of less researched but nonetheless highly impactful ones. Chapters cover military uses of AI in warfare (Ingvild Bode and Guangyu Qiao-Franco), policing and law enforcement (Mareile Kaufmann), migration and border control (Petra Molnar), judicial decision-making (Juan David Gutiérrez as well as David Mark, Tomás McInerney and John Morison), labour and employment (Robert Donoghue, Huanxin Luo, Phoebe Moore and Ekkehard Ernst), welfare services (Lyndal Sleep and Joanna Redden), social care (Grace Whitfield, James Wright and Kate Hamblin), child protection (Jenny Krutzinna), health care (Mirjam Pot and Barbara Prainsack) and urban governance (Ilia Antenucci and Fran Meissner).

NOTE

1. Of course, in the end, the consequences of the climate emergency, biodiversity loss, landscape degradation and poisoning of life from such activities, will be felt by all. But, as with the discussion on informational capitalism, the loss is first and mostly felt by those people and places who were the first objects of such appropriations and extractions.

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